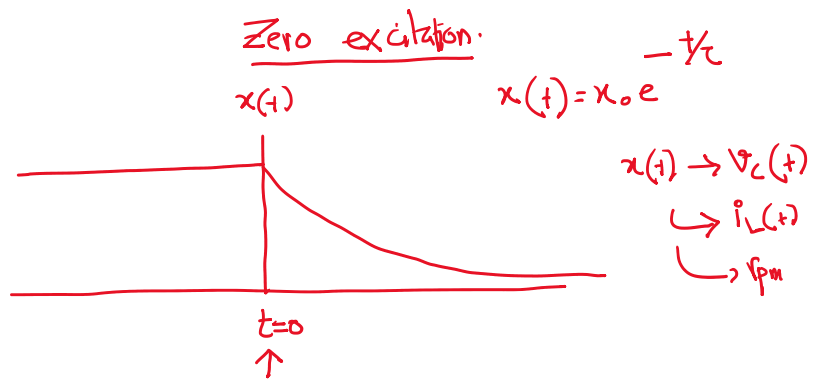


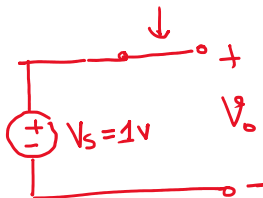
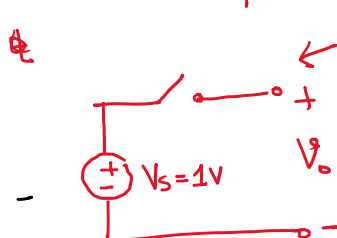
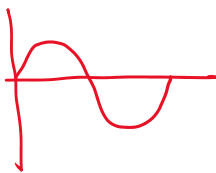


Forced Response

If the system is in standstill/steady state
 Condition for $t < 0$, at $t = 0$ instant
 it receives excitation.

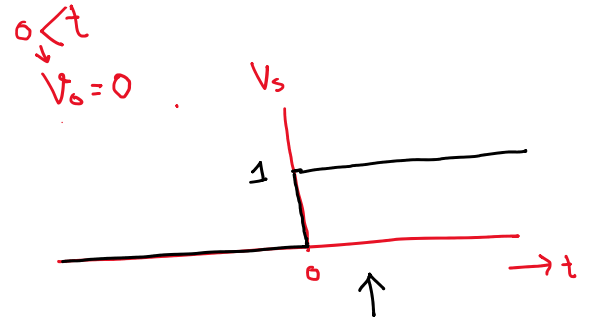
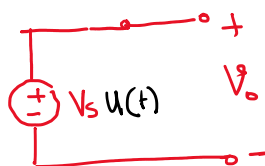
- Standstill, turn the Switch ON
- Fan is rotating at certain speed for long time (steady state), then at $t = 0$, you up the speed.

$\sin \theta$



$$u(t) = 0, \text{ for } t < 0$$

$$1, \text{ for } t \geq 0$$

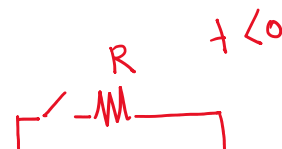


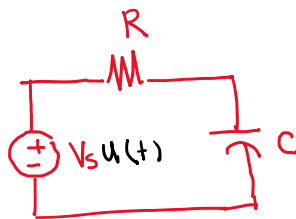
$V_0 = V_s = 1V$ Unit step function

$u(t)$ [EEE 243, Signals and system]

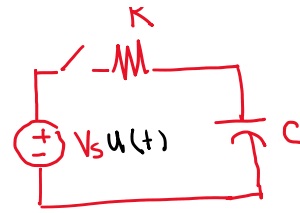
$$V_s u(t) = V_s \times 0 = 0 \text{ for } t < 0 \text{ (OFF)}$$

$$= V_s \times 1 = V_s \text{ for } t \geq 0 \text{ (ON)}$$



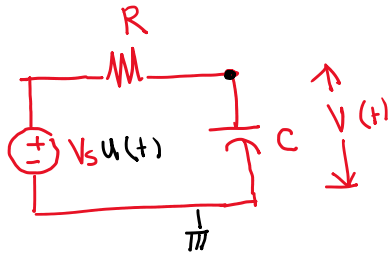
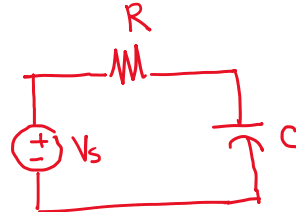


$\rightarrow$



$t > 0$

$\searrow$



$$\frac{V - V_s u(t)}{R} + C \frac{dV}{dt} = 0$$

$$\Rightarrow V - V_s u(t) + RC \frac{dV}{dt} = 0$$

$$V \equiv V(t).$$

$$t \geq 0, \\ u(t) = 1$$

$$= V - V_s + RC \frac{dV}{dt} = 0$$

$$\Rightarrow \frac{dV}{dt} + \frac{V}{RC} = \frac{V_s}{RC}$$

$$\Rightarrow \frac{dV(t)}{dt} + \frac{V(t)}{\frac{V_s}{RC}} = \frac{V_s}{RC} \Leftrightarrow \frac{dV(t)}{dt} + \frac{V(t)}{P} = \frac{Q}{P}$$

$$P = \frac{1}{RC}$$

$$Q = \frac{V_s}{RC}$$

$$\Rightarrow V(t) = \frac{Q}{P} + A e^{-Pt}$$

$$x(t) = \frac{Q}{P} + A e^{-Pt}$$

$$V(t) = \frac{\frac{V_s}{RC}}{\frac{1}{RC}} + A e^{-\frac{t}{RC}}$$

$$V(t) = V_s + A e^{-t/RC}$$

Apply Boundary Condition
at $t=0$.

$$V(0) = V_s + A e^{-\frac{0}{RC}}$$

$$V(0) = V_s + A$$

$$V(t) \text{ at } t=0$$

$$V(0) = V(0^-)$$

$$\Rightarrow A = V(0) - V_s$$

$$V(t) = V_s + [V(0) - V_s] e^{-t/RC}$$

$t \rightarrow \infty$

$$V(0) = V(0^-)$$

$$V(t) = V_s + [V(0) - V_s] e^{-t/\tau}$$

$$t \geq 0, t \rightarrow \infty \quad V(\infty) = V_s + [V(0) - V_s] e^{-\infty/\tau}$$

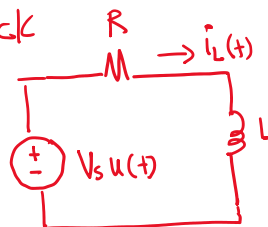
$$V_s = V(\infty)$$

$$V(t) = V(\infty) + [V(0) - V(\infty)] e^{-t/\tau} \quad \tau = RC$$

RC ckt

$$V(t) = V(\infty) + [V(0) - V(\infty)] e^{-t/\tau}$$

$$RL \text{ ckt} \quad R \quad L \quad i_L(t) \quad V_s u(t) \quad i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau} \quad \tau = \frac{L}{R}$$

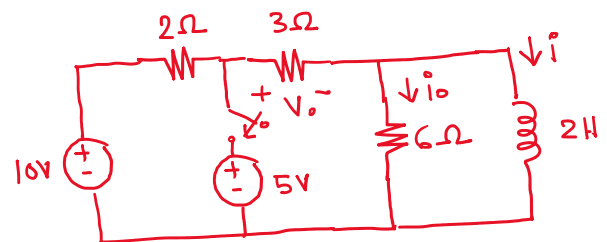


$-\infty < t < \infty$ for all time

Switch open for $t < 0$,

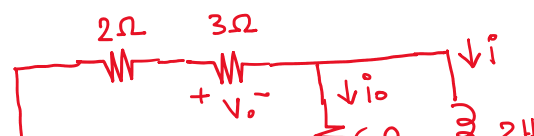
for $t \geq 0$

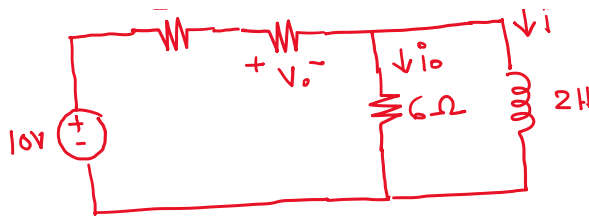
$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$



for $t < 0$, switch is open

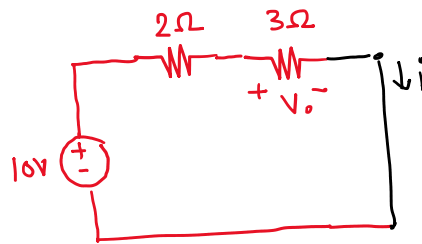
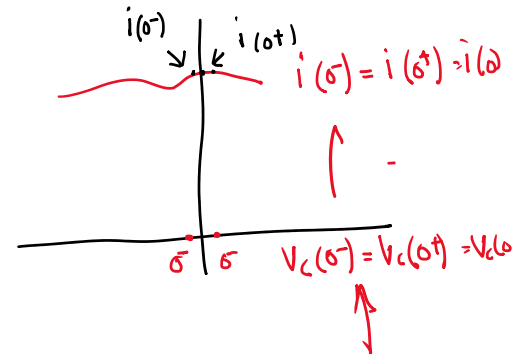
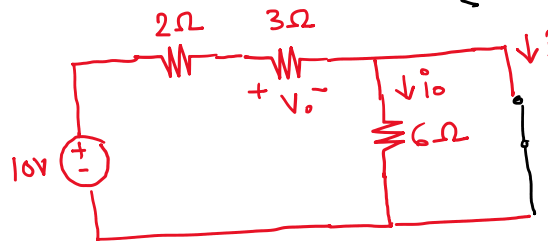
$$i(0) = i(0^-)$$





$$i(0) = i(0^-)$$

Inductor short (KT)

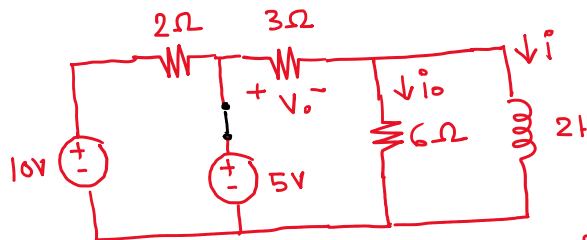


$$i(0^-) = \frac{10}{5} = 2A$$

$$i_o(0^-) = 0$$

$$V_o(0^-) = i(0^-) 3 = 6V$$

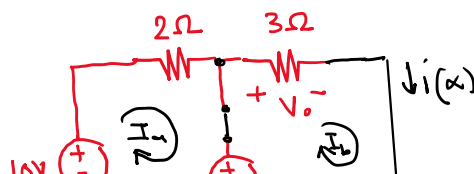
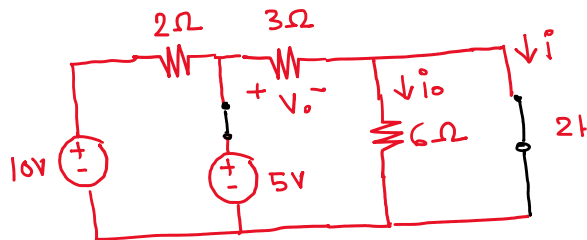
$$i(0) = i(0^-) = 2A$$



$t=0$, Switch closed

$i(\infty)$: Steady state. to DC Signal Source inductor S.C

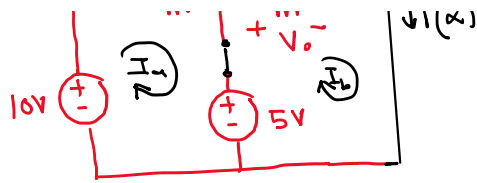
∞ : means that the switch is closed for long long time $\rightarrow$ Steady state (doesn't change with time)



$$+ 10 - 2I_a - 5 = 0$$

$$\Rightarrow I_a = 2.5A$$

$$i(\infty) = \frac{10}{2+3+6} = 1.17A$$



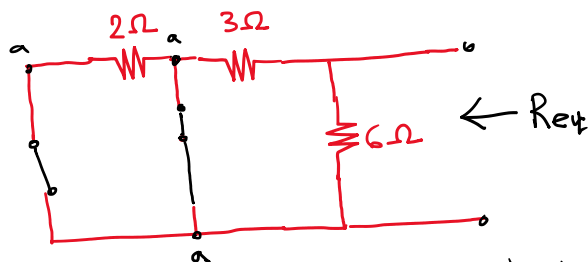
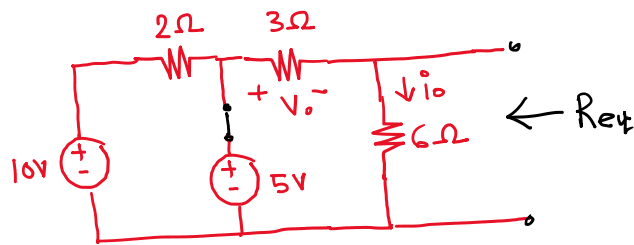
$$\Rightarrow I_a = 2.5 \text{ A}$$

$$+5 - 3I_b = 0 \Rightarrow I_b = \frac{5}{3} = 1.67 \text{ A}$$

$$i(\infty) = I_b = 1.67 \text{ A}$$

$$i_o(\infty) = 0, \quad V_o(\infty) = 3 \times 1.67 = 5 \text{ V}$$

$$\tau = \frac{L_{eq}}{R_{eq}} \quad \text{When } t > 0$$



2 Ω Resistor Shorted

$$R_{eq} = 3 \parallel 6 \Omega = \frac{18}{9} = 2 \Omega$$

$$\tau = \frac{2}{2} = 1 \text{ sec}$$

$$i(t) = i(\infty) + [i(0) - i(\infty)] e^{-t/\tau}$$

$$i(t) = 1.67 + [2 - 1.67] e^{-t/1}$$

$$i(t) = 1.67 + 0.37 e^{-t}$$

